Purpose and application of procedure: This standard operating procedure describes techniques for inoculating media used in the culture and identification of bacterial and fungal pathogens.

MASTER

Procedure:

I. **Personal Protective Equipment**
   A. Biological safety cabinet
   B. Gloves
   C. Laboratory Coat

II. **Materials needed**
   A. Bacteriological loop/needle
   B. Bunsen burner
   C. Swabs
   D. Test tube racks

III. **Media**
   A. **Plate media**
      Bismuth Sulfite Agar
      Brilliant Green Sulfa Agar (BGS)
      BUG Agar with 5% Sheep Blood (BUG)
      Campylobacter Blood-Free Selective Agar (CAMP)
      Chocolate Columbia Blood Agar (CHOC)
      Columbia Blood Agar (BA)
      Cysteine Heart Agar
      Ethyl Violet Agar (EV)
      Hektoen Enteric Agar
      Inhibitory Mold Agar (IMA)
      MacConkey Agar (MAC)
      Modified Semi-solid Rappaport-Vassiliadis Agar (MSRV)
      Mycoplasma Medium (MYCO)
      Mycosel Agar (MYC)
      Phenylethanol Agar (PEA)
      Potato Dextrose Agar (PDA)
      Sabouraud Dextrose Agar (SAB)
      Serum Agar (SER)
      Tryptose Agar (TA)
      Xylose-Lysine Agar Supplemented with Tergitol 4 (XLT4)
      Xylose Lysine Desoxycholate Agar (XLD or XLDN)
   B. **Agar tube media**
      Bile Esculin Agar

Clarks media
Columbia Blood Agar (BA)
Herrold's Egg Yolk Agar with and without Mycobactin
Leptospira Medium Semisolid Base EMJH
Leptospira Medium NVSL
Lowenstein Jensen Agar
Lysine Iron Agar (LIA)
Potato Dextrose Agar (PDA)
Sabouraud Dextrose Agar (SAB)
SIM Medium (SIM)
Simmons Citrate Agar (CIT)
Triple Sugar Iron Agar (TSI)
Tryptose Agar (TA)

C. **Liquid tube media**
Coagulase Plasma EDTA
Heart Infusion Broth (Sugar Disks)
Leptospira Medium Base EMJH
Leptospira Medium NVSL
Malonate Broth
Urea Broth

D. **Liquid media in whirlpaks or tubes**
Brain Heart Infusion Broth
Tetrathionate Broth

IV. Inoculation of media for culture

A. Culture media should be inoculated from most specimens or samples in the "well" or edge of the media plate as described for each specimen type in BAC.235 Processing Specimens.

B. Specimens are inoculated to culture media as assigned according to BAC.239 Procedure for Assigning Tests.

C. Tissue samples are inoculated to plates with no more than two samples per divided plate. Mycoplasma cultures may have up to four samples per divided plate.

D. Milk cultures may have as many as four samples per divided plate on BA. Mycoplasma cultures from milk samples may have as many as 20 samples per plate.

E. Yolk sac samples from day old chicks may have as many 2 to 3 samples per plate.

F. Samples from the same source may be pooled unless otherwise requested by a case coordinator or client.

G. Samples for Salmonella culture are pooled for composite culture in a selective enrichment media, tetrathionate broth.

V. Isolation of bacterial and/or fungal colonies on plate

A. To identify microorganisms on culture media it is necessary to isolate individual colonies. Streaking for isolation dilutes bacteria or fungi present in a sample over three to four quadrants of a plate.

B. Bacterial plate media for specimen culture or for colony re-isolation may be streaked using the quadrant method or a similar adaptation of the quadrant method for the isolation of individual bacterial colonies.

C. Fungal plate cultures from specimens such as aspirates, exudates, fluids, small biopsies, and swabs, etc. may be inoculated and streaked using the quadrant method or a similar adaption for the isolation of fungal colonies.

D. Streaking for isolation may be done with sterile disposable loops, sterile bacteriological needles, or sterile bacteriological loops.

E. If disposable loops or needles are used, heat sterilizing between streaks is not necessary.

F. An alternative to heat sterilizing between streaks to dilute bacteria on the loop or needle is to stab the agar on the plate and/or drag the loop or needle through the agar between sections.

G. The quadrant method is used as an example of isolation streaking methods, but the number of sections may be adapted as needed.

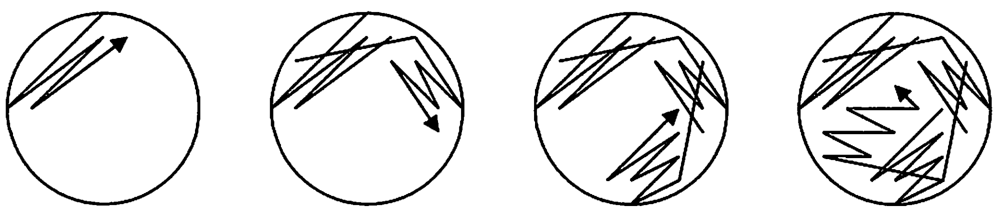
VI. Quadrant method of streaking for isolation

A. Plates are streaked for isolation after inoculation as follows (Culture plates that have been divided for multiple samples will be streaked for isolation differently as outlined in Section VII of this SOP.):

1. For the quadrant method of isolation, the “well” of the plate is the first quadrant. (This differs for divided plates with multiple samples and plates with other numbers of sections.)
    a. For specimens, the plate is inoculated as outlined in BAC.235 Procedure for Processing Specimens.
    b. For re-streaking colonies, a sterile bacteriological loop or needle is used to inoculate the plate in the “well” of the plate. The loop or needle is heat sterilized and allowed to cool before moving to the second quadrant. If using a disposable loop, stab the agar with the loop to dilute the inoculum and continue streaking to the second quadrant.
2. Streak the second quadrant of the plate by touching the loop into the first quadrant and streaking all the way across to the second quadrant, making several strokes in a zigzag motion.
3. Heat sterilize and cool the loop or drag the loop through the agar to dilute. If using a disposable loop, stab the agar with the loop to dilute the inoculum and continue streaking to the third quadrant.

4. Streak the third quadrant by touching the loop into the second quadrant and streaking into the third quadrant, making several strokes in a zigzag motion.
5. Heat sterilize and cool the loop or drag the loop through the agar to dilute. If using a disposable loop, stab the agar with the loop to dilute the inoculum and continue streaking to the fourth quadrant.
6. Streak the fourth quadrant by touching the loop into the third quadrant and streaking into the fourth quadrant, making a continuous streak in a zigzag pattern.
7. Heat sterilize and cool the loop. Discard the loop if using a disposable loop.

Example:

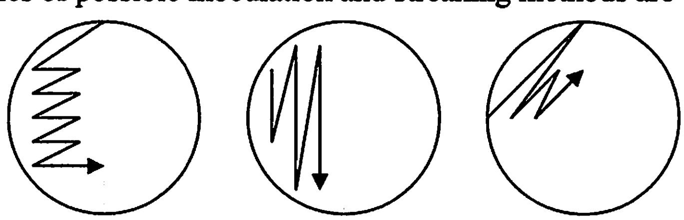


Quadrant 1 Quadrant 2 Quadrant 3 Quadrant 4

VII. Divided plates/multiple samples

A. Plates divided for culturing multiple specimens or for re-streaking multiple colonies are streaked by inoculating the plate in the "well" of each section and with a zigzag motion, streak through the entire area of the section. Some examples of possible inoculation and streaking methods are below.

Example:

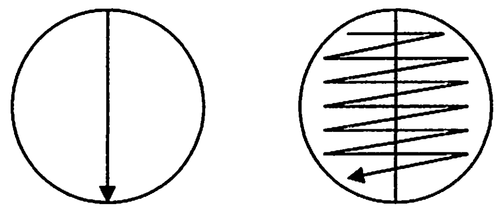


Half plate Half plate Quartered plate

VIII. Urine culture

A. Inoculating and streaking plates for urine culture is done by inserting a sterile disposable loop in the urine and then streaking once down the center of the culture plate. Next without heat sterilizing or diluting, using a zigzag motion, streak the plate across the center line from one end to the other.

Example:



Step 1 Step 2

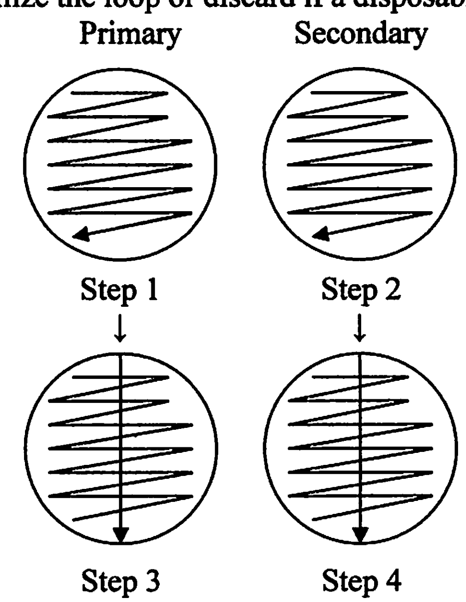
IX. Avibacterium/Haemophilus culture

A. Avibacterium/Haemophilus cultures are inoculated and streaked to Columbia Blood Agar using *Staphylococcus aureus* to provide X and V factors for growth.

B. Two Columbia Blood Agar plates, a primary and secondary, are needed for this technique.

1. The sample is inoculated to the "well" of the primary plate.
2. A sterile bacteriological loop is used to streak the plate using a lawn streak motion as shown below (step 1). Without heat sterilizing the loop or diluting in any way, move to the secondary plate and again streak using a lawn streak motion (step 2).
3. Heat sterilize the loop and allow it to cool. Discard if it is a disposable loop.
4. Touch a sterile loop to the top of a *Staphylococcus aureus* colony.
5. Inoculate both plates with the *Staphylococcus aureus* down the center by drawing a single line with the loop beginning with the primary plate (steps 3 and 4).
6. Heat sterilize the loop or discard if a disposable loop is used.

Example:

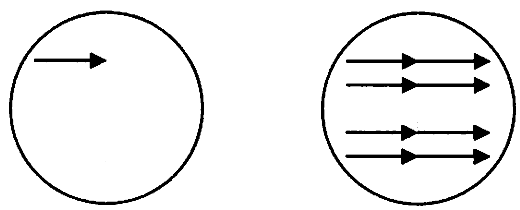


X. Mycoplasma culture

A. Mycoplasma medium is used for the culture of Mycoplasma species and is inoculated as follows:

1. Fluid specimens are inoculated to Mycoplasma media with a loop by inserting the sterile loop into the fluid and then gently dragging the loop in a single line across the surface of the media. Do not attempt to streak for isolation.
2. Swabs are inoculated to Mycoplasma media by gently rolling the swab in a single line across the surface of the media. Do not attempt to streak for isolation.
3. Below is an example of streaking Mycoplasma media for Mycoplasma culture on a divided plate.

Example:



Single Sample Multiple Samples

XI. Modified Semi-solid Rappaport-Vassiliadis Agar

A. Modified Semi-solid Rappaport-Vassiliadis Agar is inoculated as directed in BAC.255 Culture of Salmonella from Environmental and Meconium Samples.

XII. Inoculating fungal plate media

A. Specimens are placed on fungal growth plates as outlined in BAC.235 Procedure for Processing Specimens.

B. Re-streaking or subbing organism for growth on fungal media varies by organism.

1. Yeast colonies may be re-streaked for isolation using the inoculation and streaking procedures as outlined in Section V-VII of this SOP.
2. Subbing other types of fungi for isolation and/or identification consists of asceptically transferring a portion of a colony with a sterile loop or needle to a new plate.

XIII. Inoculating agar tube media for culture and/or biochemical testing

A. Tube media may be inoculated using sterile disposable loops, bacteriological loops, or bacteriological needles. Inoculating needles are preferred for inoculating SIM Medium for detecting motility.

B. Biochemical tube media should only be inoculated with growth from pure cultures or well isolated colonies.

C. The butt of a tube agar media is the lower portion of the tube below the surface of the agar. The surface of the agar is sometimes referred to as the slant.

D. Triple Sugar Iron Agar and Lysine Iron Agar are inoculated using a stab-streak method:

1. Heat sterilize the inoculating loop or needle and allow it to cool. No heat sterilization is necessary if a disposable loop or needle is used.
2. Touch the top of a colony (if pure culture any area may be used) with a sterile inoculating loop or needle.
3. Uncap the agar tube.
4. Insert the loop or needle into the butt of the agar and then streak the slant using a zigzag motion like the lawn streak method.
5. Loosely cap the agar tube.
6. Heat sterilize the loop or needle or discard the disposable loop or needle.

E. Bile Esculin Agar, Simmons Citrate Agar, and TA are inoculated by the following streak method:

1. Heat sterilize the inoculating loop or needle and allow it to cool. No heat sterilization is necessary if a disposable loop or needle is used.
2. Touch the top of a colony (if pure culture any area may be used) with a sterile inoculating loop or needle.
3. Uncap the agar tube.
4. Streak the slant using a zigzag motion similar to the lawn streak method.
5. Loosely cap the agar tube.
6. Heat sterilize the loop or needle or discard the disposable loop or needle.

F. SIM Medium is inoculated using a stab only method:

1. Heat sterilize the inoculating needle and allow it to cool. No heat sterilization is necessary if a disposable needle is used.
2. Touch the top of a colony (if pure culture any area may be used) with a sterile inoculating needle.
3. Uncap the agar tube.
4. Insert the needle into the butt of the agar.
5. Loosely cap the agar tube.
6. Heat sterilize the needle or discard the disposable needle.

G. Herrold's Egg Yolk Agar with and without Mycobactin is inoculated and incubated as outlined in BAC.234 Procedure for Johne's Culture.

H. Clarks media is inoculated as outlined in BAC.014 Procedure for Preparation of Clarks Media and Culture of *Campylobacter fetus*.

I. Lowenstein Jensen Agar is inoculated and incubated as outline in BAC.266 Procedure for Culture of Mycobacterium spp. from Body Fluids, Tissues, and Swabs.

XIV. Inoculating liquid media

A. Liquid tube media to include Coagulase, Heart Infusion Broth, Malonate Broth, and Urea Broth are inoculated using the following method:
   1. Heat sterilize the inoculating loop or needle and allow it to cool. No heat sterilization is necessary if a disposable loop or needle is used.
   2. Touch the top of a colony (if pure culture any area may be used) with a sterile inoculating loop or needle.
   3. Uncap the medium tube.
   4. Insert the loop or needle into the broth and mix to remove the inoculum from the loop or needle. (For Heart Infusion Broth if a sugar disk is in the tube, push the disk into the media with the loop or needle while inoculating the media.)
   5. Loosely cap the medium tube.
   6. Heat sterilize the loop or needle or discard the disposable loop or needle.

B. Brain Heart Infusion Broth is inoculated as follows (For Listeria culture from specimens using Brain Heart Infusion Broth, see BAC.213.)
   1. Heat sterilize the inoculating loop or needle and allow it to cool. No heat sterilization is necessary if a disposable loop or needle is used. Sterile swabs may be used for inoculating liquid medium also.
   2. Touch the top of a colony (if pure culture any area may be used) with a sterile inoculating loop, needle, or swab.
   3. Insert the loop or needle into the broth and mix to remove the inoculum from the loop or needle. If a swab is used, insert the swab into the broth and mix to remove the inoculum. Leave the swab in the liquid medium for incubation.
   4. Heat sterilize the loop or needle or discard the disposable loop or needle.

C. Tetrathionate Broth is inoculated as follows:
   1. Specimens are cultured in Tetrathionate Broth, selective enrichment broth for Salmonella culture, as outlined in BAC.235.
   2. Tetrathionate Broth is inoculated with environmental samples from poultry samples as outlined in BAC.255 Culture of Salmonella from Environmental and Meconium Samples.
   3. Culture of poultry samples for Salmonella pullorum is outlined in BAC.261 Delayed Secondary Enrichment from Poultry Tissue for Salmonella Isolation.
   4. Tetrathionate broth is inoculated to Brilliant Green Sulfa Agar and XLT4 or XLDN after incubation. Plates are streaked for isolation as outlined above in section VI or VII of this SOP.

D. Leptospira media is inoculated with a sterile pipette as outlined in BAC.212 Procedure for the Leptospira Microagglutination Test.

XV. Incubation of media

The following is a list of typical incubation times, temperatures, and atmospheric requirements for the initial isolation of bacterial and fungal pathogens from samples and for biochemical identification media. Subcultures of organisms from original cultures may require different incubation times. Adaptations may be necessary for some bacterial and fungal isolates which require longer incubation times, changes to temperature or alterations to the incubation atmosphere.

Aerobic and CO_2 cultures

Culture media	Atmosphere	Temperature	Time
Biolog Universal Growth	10% CO_2	~37°C	18-48 hours
Bismuth Sulfite Agar	Aerobic	~37°C	18-24 hours (1st read)
BG Sulfa Agar	Aerobic	~37°C	18-24 hours
Chocolate Agar	10% CO_2	~37°C	18-24 hours (1st Read)
Columbia Blood Agar	10% CO_2	~37°C	18-24 hours (1st Read)
Cysteine Heart Agar	Aerobic	~37°C	48 hours (1st Read)
Ethyl Violet Agar	10% CO_2	~37°C	7-10 days
Hektoen Enteric Agar	Aerobic	~37°C	18-24 hours
MacConkey Agar	Aerobic	~37°C	18-24 hours
Modified Semi-solid - Rappaport-Vassiliadis Agar	Aerobic	~42°C	48 hours
Mycoplasma Agar	10% CO_2	~37°C	7-10 days
Phenylethanol Agar	10% CO_2	~37°C	18-24 hours (1st Read)
Serum Agar	10% CO_2	~37°C	7-10 days
Tetrathionate Broth	Aerobic	~37°C	18-24 hours
Tryptose Agar	Aerobic	~37°C	18-24 hours
XLD or XLDN	Aerobic	~37°C	18-24 hours
XLT4	Aerobic	~37°C	18-24 hours

Biochemical media	Atmosphere	Temperature	Time
Bile Esculin Agar	Aerobic	~37°C	18-24 hours
Coagulase Plasma	Aerobic	~37°C	18-24 hours
Heart Infusion Broth	Aerobic	~37°C	18-24 hours
Lysine Iron Agar	Aerobic	~37°C	18-24 hours
Malonate Broth	Aerobic	~37°C	18-24 hours
SIM Medium	Aerobic	~37°C	18-24 hours
Simmons Citrate Agar	Aerobic	~37°C	18-24 hours
Triple Sugar Iron Agar	Aerobic	~37°C	18-24 hours
Urea Broth	Aerobic	~37°C	18-24 hours

Anaerobic cultures

Culture media	Atmosphere	Temperature	Time
Columbia Blood Agar	Anaerobic	~37°C	48 hours
Phenylethanol Agar	Anaerobic	~37°C	48 hours

Microaerophilic cultures

Culture media	Atmosphere	Temperature	Time
Charcoal Agar	Microaero.	~42°C	2-3 days (C. jejuni)
Columbia Blood Agar	Microaero.	~42°C	2-3 days (C. jejuni)
Columbia Blood Agar	Microaero.	~37°C	5-7 days (C. fetus)

Fungal cultures

Culture media	Atmosphere	Temperature	Time
Inhibitory Mold Agar	Aerobic	Room	4 to 6 weeks
Mycosel Agar	Aerobic	Room	4 to 6 weeks
Potato Dextrose Agar	Aerobic	Room	4 to 6 weeks
Sabouraud Dextrose Agar	Aerobic	Room	4 to 6 weeks

XVI. Disposal

See BAC.232 Procedure for the Disposal of Laboratory Waste.

Associated Forms or Files:

N/A

Associated SOPs:

BAC.005 Procedure for Preparation of Brain Heart Infusion Broth
BAC.006 Procedure for Preparation of BG Sulfa Agar
BAC.008 Procedure for Preparation of Campylobacter Blood Free Selective Agar
BAC.010 Procedure for the Preparation of Chocolate Columbia Blood Agar
BAC.012 Procedure for Preparation of Coagulase Plasma EDTA
BAC.013 Procedure for Preparation of Columbia Blood Agar
BAC.014 Procedure for Preparation of Clarks Media and Culture of *Campylobacter fetus*
BAC.015 Procedure for Preparation of EV Agar
BAC.018 Procedure for the Preparation of Ehrlich's Reagent
BAC.023 Procedure for the Preparation of Heart Infusion Broth
BAC.024 Procedure for the Preparation of Inhibitory Mold Agar
BAC.026 Procedure for the Preparation of Leptospira Medium Base EMJH
BAC.028 Procedure for Preparation of Lysine Iron Agar
BAC.029 Procedure for Preparation of MacConkey Agar
BAC.030 Procedure for Preparation of Malonate Broth
BAC.032 Procedure for Preparation of Mycosel Agar

BAC.033 Procedure for preparation of Modified Semi-solid Rappaport-Vassiliadis Agar
BAC.034 Procedure for the Preparation of Phenylethanol Agar with 5% Sheep Blood
BAC.037 Procedure for the Preparation of Potato Dextrose Agar
BAC.038 Procedure for the Preparation of Sabouraud Dextrose Agar
BAC.039 Procedure for the Preparation of 0.85% Saline Solution
BAC.040 Procedure for Preparation of Serum Agar
BAC.041 Procedure for the Preparation of SIM Medium
BAC.042 Procedure for the Preparation of Simmons Citrate Agar
BAC.043 Procedure for the Preparation of Tetrathionate Broth Base
BAC.044 Procedure for the Preparation of Triple Sugar Iron Agar
BAC.045 Procedure for the Preparation of Tryptose Agar
BAC.046 Procedure for the Preparation of Urea Broth
BAC.047 Procedure for the Preparation of Xylose-Lysine Agar Supplemented with Tergitol 4
BAC.048 Procedure for the Preparation of BUG Agar with 5% Sheep Blood
BAC.052 Procedure for the Preparation of Bile Esculin Agar
BAC.053 Procedure for the Preparation of XLD Agar
BAC.054 Procedure for the Preparation of Bismuth Sulfite Agar
BAC.055 Procedure for the Preparation of Hektoen Enteric Agar
BAC.211 Procedure for Kirby Bauer Disk Diffusion Antibiotic Susceptibility Test
BAC.212 Procedure for the Leptospira Microagglutination Test
BAC.213 Procedure for Listeria Cold Enrichment
BAC.232 Procedure for the Disposal of Laboratory Waste
BAC.234 Procedure for Johne's Culture
BAC.235 Procedure for the Processing of Specimens
BAC.239 Procedure for Assigning Tests
BAC.240 Procedure for Anaerobic and Microaerophilic Cultures
BAC.255 Culture of Salmonella from Environmental and Meconium Samples
BAC.261 Delayed Secondary Enrichment from Poultry Tissue for Salmonella Isolation
BAC.264 Procedure for Detection of Salmonella enteritidis from Eggs Using
the Romer Labs RapidChek Test System
BAC.266 Procedure for Culture of Mycobacterium spp. from Body Fluids, Tissues, and Swabs

Revisions:
III-A, XIV-C-4, XV- Added XLDN.

References:
Clinical Veterinary Microbiology, 2nd Edition, 2013.
Veterinary Microbiology, 3rd Ed., 2013.
Essentials of Veterinary Bacteriology and Mycology, 6th Ed., 2004.
Diagnostic Procedures in Veterinary Bacteriology and Mycology, 5th Ed., 1990.
Bacteriological Analytical Manual, Chapter 5 Salmonella, 02/2020, FDA.

Quality Control:
N/A

Policy Implementation steps:
N/A